Targeted Pitching Biomechanics in Training
Executive Summary – Spring 2026

Capstone Team

Maria Armaganian, Kalen McNamara, Quincy Steidle, Olivia Valenti

Advisor

Professor B. Kris Jaeger-Helton

Real-time biomechanical feedback is essential for baseball pitcher development and injury prevention, yet current motion capture systems remain inaccessible outside elite training and performance environments due to cost and complexity. This project develops a portable system that measures pitching mechanics using inertial measurement units (IMUs) and delivers objective feedback in real time. There is no known system in existence that accomplishes what this new system does. Due to the cutting-edge nature of this project, disclosure of certain elements of the project is limited by a confidentiality agreement to protect the competitive advantage of this project's sponsor, the Boston Red Sox. The new system is designed to measure lead knee extension from foot plant to ball release, a key metric linked to both performance and injury risk. The hybrid IE/ME Capstone team (the Team) has built a multi-component system consisting of (1) a real-time sensor pipeline, and (2) integrated event detection through supplemental streaming sources. They also researched and developed a wearable sleeve configuration for consistent sensor placement on the lower limb. Preliminary results show accurate detection of key pitching events and meaningful lower-body kinematics during live throwing. Ongoing work focuses on improving calibration, filtering, and validation. This system aims to make biomechanical feedback more accessible, enabling athletes and coaches to use real-time data to improve performance and reduce injury risk.

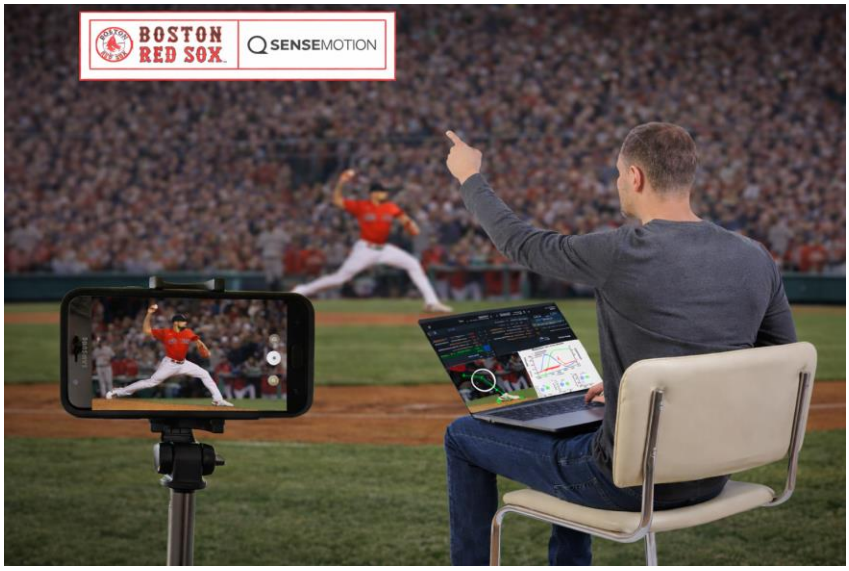


Figure 1. With this new system, key feedback can be provided to players nearly instantaneously.

For more information, please contact bk.jaeger-helton@northeastern.edu

Need

Biomechanical analysis has become essential for improving performance and reducing injury risk in modern baseball. Organizations rely on advanced motion capture systems to analyze mechanics, but these technologies are expensive, difficult to deploy, and constrained to certain environments. As a result, athletes lack access to real-time biomechanical feedback in training environments that is both reliable and actionable. This gap creates a major limitation in player development. Without immediate objective feedback, athletes must rely on subjective coaching cues or delayed feedback loops, which can delay skill acquisition and reinforce inefficient or unsafe movement patterns. This project addresses this problem by developing an accessible, portable system that provides real-time biomechanical feedback, with a focus on lead knee extension between foot plant and ball release, a key metric strongly linked to pitch velocity and injury risk mitigation.

Background and Prior Work

Biomechanical analysis in baseball pitching centers on understanding how body segments move in sequence to generate velocity while minimizing injury risk. A key concept is the kinetic chain, the coordinated transfer of energy from the legs through the trunk and into the throwing arm. Within this chain, lead knee extension, the straightening of the front leg at the knee between foot plant and ball release, plays a critical role. As the lead foot braces against the ground, it creates a stable pivot that allows rotational energy to transfer from the lower extremity to the torso. Figure 2 shows the change in lead knee angle between foot plant and ball release in a pitching delivery. Research has shown that greater lead knee extension during this phase is strongly associated with higher pitch velocity, making it a valuable and specific target for training intervention.

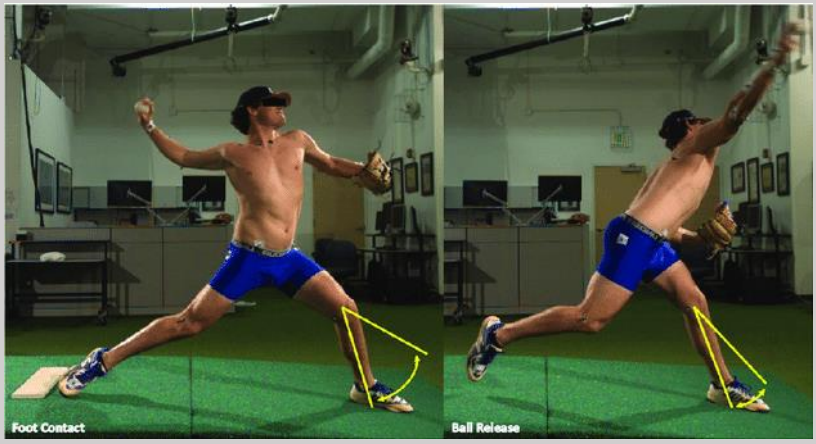


Figure 2. Lead Knee Extension during the pitching delivery is a key metric for performance and injury biomechanical analysis.

Measuring this metric accurately in real time, however, is technically challenging. The pitching motion is one of the fastest movements in sport, with end-point velocity of the upper extremity reaching angular velocities exceeding 7,000 degrees per second. Capturing lower-body kinematics, though not as great in magnitude as upper-body kinematics, requires sensors with high sampling rates and processing pipelines with minimal latency.

The current landscape of pitching biomechanics tools spans a wide range of cost, accuracy, and accessibility. At the high end, markerless optical systems such as KinaTrax and Hawk-Eye use synchronized multi-camera arrays and computer vision algorithms to reconstruct full-body 3D kinematics with exceptional precision. These systems operate at high frame rates and can deliver detailed mechanical profiles suitable for professional decision-making. However, these systems have limited accessibility due to their installation in permanent, expensive Major League stadiums and select training complexes as well as the requirement of time-consuming third-party processing before actionable insight can be extracted.

On the lower-cost end of the spectrum, camera-based pose estimation tools such as PitchAI have demonstrated that single-camera markerless approaches can estimate key kinematic variables with reasonable agreement to gold-standard systems. While the existing systems are promising for general screening, these tools depend on camera angles, lighting, and resolution, and their accuracy can degrade when analyzing rapid, complex movements where limb occlusion is common.

Wearable—or attachable—inertial measurement units (IMUs) offer a portable alternative. IMUs, such as those shown in Figure 3, combine accelerometers, gyroscopes, and magnetometers to estimate segment orientation and movement at sampling rates typically between 100 and 1000 Hz, making them viable for field deployment. However, IMU-based measurement introduces its own set of challenges. Soft tissue artifacts, where the sensor shifts relative to the underlying bone during explosive movements, can compromise or even corrupt angle estimates. Gyroscope drift accumulates over time, degrading accuracy without calibrated periodic correction. Sensor placement and alignment must be precise and repeatable, and magnetic interference from metal structures that are common in training environments can distort heading estimates. Existing wireless IMU platforms designed for high-speed athletics address some of these issues but often exceed practical cost thresholds or lack the ability to stream data from multiple sensors simultaneously in real time.

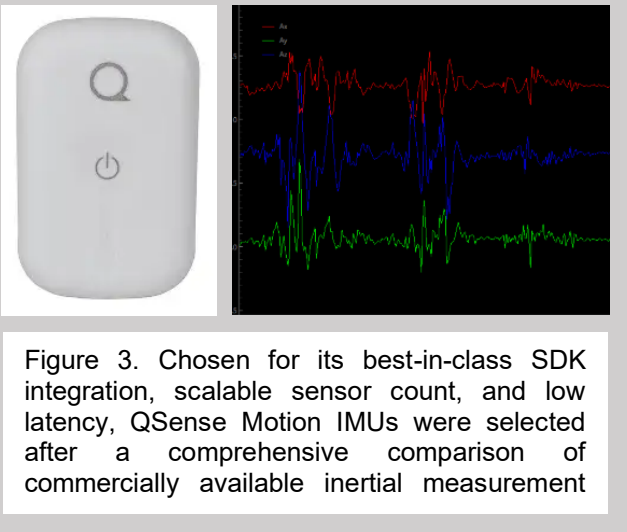


Figure 3. Chosen for its best-in-class SDK integration, scalable sensor count, and low latency, QSense Motion IMUs were selected after a comprehensive comparison of commercially available inertial measurement

In summary, existing solutions cluster at two extremes: (1) high-accuracy systems that are inaccessible to most users, and (2) low-cost tools that sacrifice the precision or usability needed for real-time biomechanical feedback. No current product or apparatus provides an affordable, portable system capable of delivering accurate, low-latency lead knee extension measurements during live bullpen sessions. Our work targets this gap directly, building a wearable IMU-based system designed to balance measurement fidelity, real-time performance, and practical usability in training environments.

Solution Approach

This project addresses the challenge of capturing and analyzing baseball pitching biomechanics in real-world environments, where traditional laboratory systems are not practical. The Team's approach was to develop a portable, field-deployable system capable of collecting high-frequency motion data and transforming it into meaningful biomechanical metrics in real time. At its core, the pipeline streams 400 Hz data from dual IMU sensors through a Bluetooth Low Energy (BLE) interface, a method of low-power, short-range wireless communication. The pipeline then converts this data into quaternions, a mathematical representation of three-dimensional angles using four-dimensional imaginary numbers, and derives key metrics through angle calculation and event detection.

The Team designed the system as a modular software pipeline to support reliable operation outside controlled environments. To achieve this, the Team implemented guardrails that enforce stable streaming configurations and prevent invalid conditions from degrading data quality. The data pipeline processes incoming sensor data through custom parsing logic, monitors stream integrity, and maintains synchronized data across multiple sensors. Built-in diagnostics track missing samples, timing irregularities, and synchronization issues, allowing system performance to be measured and validated rather than assumed.
To support practical field use, the Team developed a complete software ecosystem including a Python-based processing pipeline, a custom interface using PyQt software packages in Python for guided data collection, and a session-based storage architecture that records all trial metadata, sensor mappings, and validation outputs. The interface guides users through trial setup and execution while enforcing readiness checks and quality-gated recording. A supplemental streaming application, developed in Swift—a general-purpose, open-source programming language—was integrated to provide synchronized external timing inputs within the same workflow.

The pipeline was designed with extensibility in mind, allowing additional biomechanical metrics to be incorporated through modular event input handling and validation components. Key milestones in this approach included establishing stable dual-sensor streaming, implementing quality-gated trial capture, and developing a complete field-ready data collection workflow.



Figure 4. Custom compression sleeve with sensor pockets, designed to standardize IMU placement and maintain consistent alignment during dynamic movement.

To improve real-time data capture and reliability in dynamic field conditions, the Team developed a wearable compression sleeve (shown in Figure 4) with integrated sensor pockets to standardize sensor placement on the thigh and shank, to ensure consistent anatomical alignment across trials. While the original system relied on externally mounted straps provided by the sensor manufacturer, these were prone to shifting and inconsistent positioning during dynamic movement. To address this, a custom sleeve was designed from scratch using lightweight, four-

way stretch compression materials (polyester/elastane blend) commonly used in athletic performance wear. This material maintains continuous circumferential contact with the limb, allowing the sensors to remain securely anchored to the body while accommodating the full range of motion. The integrated pocket design was inspired by existing athlete monitoring systems, such as CATAPULT, and provides a more stable and repeatable mounting method by reducing soft tissue artifacts and minimizing relative motion between the sensor and the underlying segment. This improvement directly enhances signal stability and overall data reliability during high-speed pitching.

Results

This project successfully delivered a high-performance IMU biomechanics pipeline with repeatable high-rate streaming, automated quality controls, and end-to-end evidence generation suitable for technical review. The Team demonstrated stable dual-sensor operation at both 200 Hz and 400 Hz, completing four formal 50-run campaigns with Quality Gate pass rates of 96%-100%, confirming strong reliability under controlled operating conditions.

Through iterative development and testing, the Team established verified high-rate streaming of approximately 400 Hz in controlled test conditions. At 400 Hz, the nominal sample period is 2.5ms, and the system consistently met high pass-rate quality criteria in batch testing. Early reliability challenges were addressed by correcting link-capacity handling, improving parsing and ingest behavior, and introducing system guardrails that prevent unstable configurations. These improvements significantly reduced packet loss and improved overall system stability.

Rather than relying on ideal conditions, the system was engineered to operate under real-world variability. Stream health diagnostics and quality gates (shown in Figure 5) were introduced to detect synchronization issues, quantify missing data, and enforce objective pass and fail criteria for each trial. The system also incorporates safeguards such as missing-sample tracking, synchronization monitoring, and controlled fallback behavior to maintain usable outputs when conditions degrade.

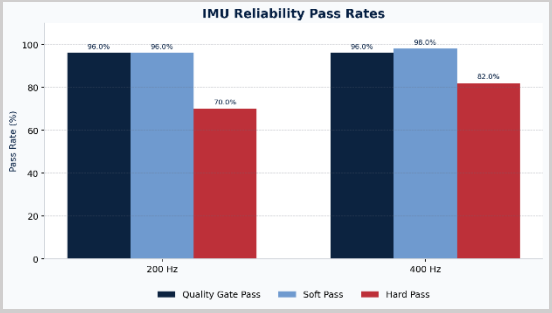


Figure 5. Reliability pass percentages across 50 trials per rate. Both rates maintained high overall pass performance.

Validation results demonstrated consistent performance across multiple test scenarios, with representative validation blocks achieving full pass rates, including endurance-style runs. Field trials also showed measurable progress in alignment quality and transport stability, alongside successful ingest and verification workflows. Each session produces complete artifacts, including metadata and validation summaries, enabling reproducibility, efficient debugging, and confidence in downstream analysis. These results demonstrate that the system is technically functional, quantitatively instrumented, and strongly positioned for final deployment in real-world environments.

Summary and Impact

This project establishes a scalable and field-ready solution for capturing and analyzing biomechanical data, bridging the gap between controlled testing environments and real-world athletic performance. By transforming raw sensor data into validated, structured outputs within a guided workflow, the system enables consistent and efficient data collection while reducing reliance on manual processes and subjective interpretation. The inclusion of quality gating and session-level validation ensures that only reliable data is used for analysis, increasing confidence for both engineers and end users. Figure 6 shows a testing setup mimicking a typical training environment, with sensor capture on a pitcher streaming to the analysis pipeline on a laptop.



Figure 6. Pipeline testing with the compression sleeve prototype using a modified mound on Carter Field near Northeastern University campus.

Beyond its immediate application, the system provides a strong foundation for future development. Its modular architecture and event-driven design allow for the integration of additional metrics and data sources without requiring significant redesign. The project demonstrates an engineering approach centered on instrumentation and validation, where system reliability is measured, enforced, and continuously improved. This reinforces the importance of designing systems that perform reliably under real-world conditions.

The work also creates a repeatable framework for future teams and stakeholders, combining real-time data capture, validation, and structured output into a single workflow. Next steps include expanding the system to additional biomechanical metrics and continuing to refine synchronization and robustness under a wider range of field conditions.